

Inferring Stratospheric Aerosol Injection Climate Response Inequality

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October 8, 2026

Abstract

Cooling the climate with stratospheric aerosol injection (SAI) would imperfectly substitute for lowering global temperatures by reducing atmospheric carbon dioxide (CO₂): some regions will cool more or less in an SAI and CO₂-forced climate than in a CO₂-only forced climate. This heterogeneity in the regional response to SAI is not well-constrained, but could be better constrained during an SAI deployment using observations of the climate response to SAI. Here, we formulate a hierarchy of models to understand the rate at which uncertainty in the inequality of the regional response to SAI – summarized by the SAI “angle parameter” – would decline during an SAI deployment. We identify three factors that influence learning about the angle parameter: (1) the SAI forcing magnitude, (2) the CO₂ forcing magnitude, and (3) the temporal covariance between the two forcings. We find that higher SAI signal accelerates learning, while higher CO₂ forcing and temporal covariance between the SAI and CO₂ forcings dampen learning. Numerical simulations suggest that the impact of the CO₂ forcing magnitude and covariance structure between SAI and CO₂ forcings on angle parameter uncertainty resolution is larger than the impact of the SAI forcing magnitude. This indicates that limiting CO₂ emissions and slowly ramping up SAI would result in the most favorable conditions for detecting inequalities in SAI benefits. We close by discussing how these results could influence coupled choices about CO₂ mitigation and SAI policy, as well as SAI scenario design.

1 Introduction

Stratospheric aerosol injection (SAI), if ever deployed at the global scale, would likely be one of the most closely monitored environmental interventions in history. This is because risks associated with SAI and its impact on global and regional climate are uncertain, and while global and regional cooling benefits are nearly certain to manifest (Irvine et al., 2019; Irvine and Keith, 2020; Fan et al., 2021; Harding et al., 2024), some regions could benefit more or less than others relative to the global average. Indeed, among the many practical concerns that would surround its deployment (Tang and Kemp, 2021; Hack et al., 2025), one concern is that the benefits of SAI could be unequally distributed, and an unequal distribution of benefits could undermine support for the program.

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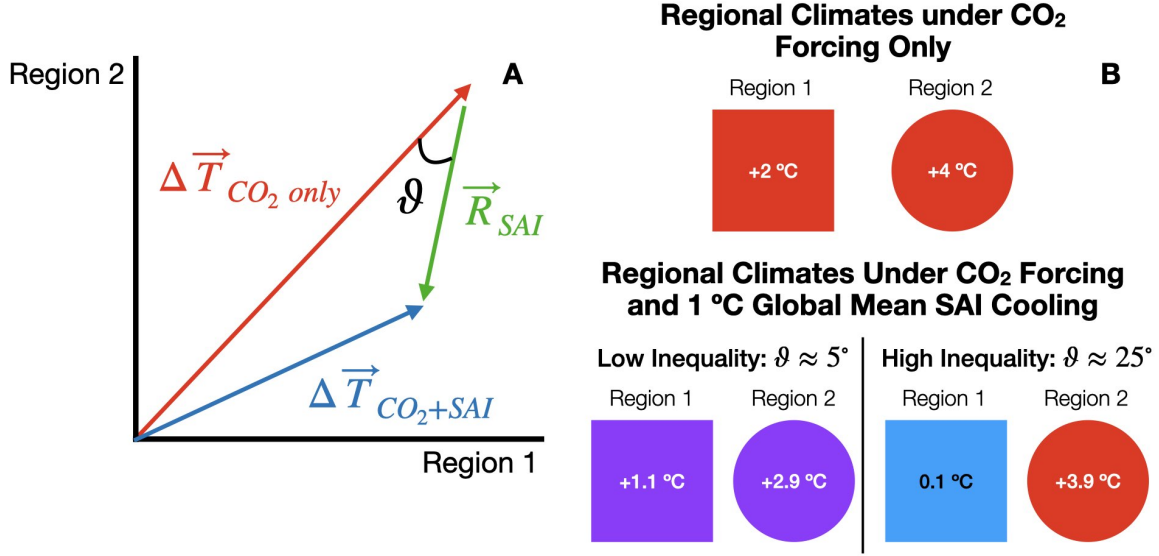


Figure 1: SAI Angle Parameter Schematic. **A** shows the SAI angle parameter ϑ as defined in the residual vector space of the regional climate response. The red arrow is the CO₂-only response, $\Delta \vec{T}_{CO_2 \text{ only}}$, the blue arrow is the CO₂+SAI response, $\Delta \vec{T}_{CO_2+SAI}$, and the green arrow is the residual vector, $\vec{R}_{SAI} = \Delta \vec{T}_{CO_2+SAI} - \Delta \vec{T}_{CO_2 \text{ only}}$. **B** shows a schematic breakdown of high and low ϑ scenarios using the two region case assuming each region has the same area. Temperatures are anomalies relative to the preindustrial era. Each panel is for illustrative purposes only.

To quantify the inequality characteristics of the SAI climate response, [Moreno-Cruz et al. \(2012\)](#) developed a simple model that summarizes the differences in response between a climate forced with CO₂ and SAI relative to the CO₂-only response. The key output of this model is the SAI “angle parameter”, ϑ , where a high (low, resp.) value of ϑ indicates an unequal (equal, resp.) response.

In Figure 1, we provide a conceptual diagram to explain the angle parameter. The central idea is to compute the climate response in a set of regions (simplified to two in the diagram) under CO₂-only forcing and CO₂ and SAI forcing. The vector of regional differences between the CO₂-only and CO₂+SAI response forms a residual vector, given by $\vec{R}_{SAI}$ in Fig. 1A. The angle between the CO₂-only response vector and the residual vector defines ϑ . One can immediately see that $\vartheta = 0^\circ$ implies that SAI perfectly offsets the CO₂ response, while $\vartheta > 0^\circ$ implies an imperfect offset. Fig. 1B provides a quantitative illustration.

This model is convenient as it extends to any number of regions and can be adapted for any climate variable that varies approximately linearly with forcing ([Moreno-Cruz et al., 2012](#)). When this approximation holds, global climate damages will be proportional to the SAI angle parameter, making it a measure of the inequality of benefits of a given SAI program. Indeed, ϑ has been utilized in climate damage functions in integrated assessment models that include SAI ([Belaia et al., 2021](#)); using the damage parameterization of [Belaia et al. \(2021\)](#) (see their Eqn. (2)), offsetting half of the radiative forcing from CO₂ with SAI leads to overall climate damages that are $\approx 43\%$ larger when $\vartheta = 35^\circ$ compared to when $\vartheta = 5^\circ$.

Because of the value of information discovery in decision-making (Dixit and Pindyck, 1994), the extent to which the SAI angle parameter can be determined could lead to substantial cost savings (or additional expenses) for policymakers. Previous work in the economics literature has quantified the value of information about the climate feedback parameter (and by extension, the equilibrium climate sensitivity) in formulating greenhouse gas abatement policy (Kelly and Kolstad, 1999; Kelly and Tan, 2015; Bauer et al., 2025a). Harding et al. (2023) carry out a similar calculation for learning about the SAI angle parameter, however learning happens instantaneously in their model rather than evolving over time in response to sequentially assimilated observations. Kelly et al. (2024) considers how jointly learning the global climate feedback parameter and SAI forcing efficacy impacts economic decision-making, finding that increasing the amount of SAI in order to narrow uncertainty is never justified, but does not consider regional impacts.

Despite the value of information about the angle parameter demonstrated in Harding et al. (2023), little attention has been paid to how quickly one could determine the angle parameter during a potential deployment of SAI. Previous work has leveraged global climate model simulations to explore the extent to which small-scale SAI deployments could reveal information about the global climate response (MacMynowski et al., 2011), optimal methods of distinguishing the SAI climate response from that of CO₂ (Lo et al., 2016, 2018), and the timescale of learning about the SAI climate response (MacMartin et al., 2019). These studies are largely numerical and lack a parsimonious explanation for why learning is fast or slow. One exception to this is Dong et al. (2026), who use explainable artificial intelligence to highlight the role of the seasonal cycle of interannual variability and shifts and modulations of the intertropical convergence zone strength to explain why distinguishing the precipitation response to SAI is more difficult than the temperature response to SAI.

A shared limitation of the prior literature (with the exception of Kelly et al. (2024)) is that each study leverages a single climate model (or an emulator trained on a particular dataset), which implies that the global and regional CO₂ responses are implicitly taken to be known: there is only one value of equilibrium climate sensitivity per climate model. Therefore, detecting the SAI response in this context is done in the presence of a known (if noisy) background field. If the SAI response was to be inferred from an observational record during deployment, however, the CO₂-driven and SAI-driven climate responses would have to be jointly estimated: the true value of Earth’s equilibrium climate sensitivity (ECS) is *not* known prior to estimation, which would make estimation more challenging.

In this paper, we build a hierarchy of models to understand the rate at which the SAI angle parameter could be inferred during a potential deployment. We begin with a conceptual, two parameter model to demonstrate the key controls on the rate of uncertainty resolution. We show that there are three knobs that control the rate of uncertainty resolution: (1) the size of the SAI-forced signal, (2) the size of the CO₂-forced signal, and (3) the temporal correlation between the SAI and CO₂ forcing signatures. We find that more SAI cooling increases learning, while higher CO₂ forcing and temporal correlation between the SAI and CO₂ forcings limit learning. We show that, in the conceptual model, near-term learning is primarily driven by the size of the SAI signal, while long-term learning is driven by the CO₂ forcing size and the covariance structure between the CO₂ and SAI forcings.

To test these conclusions, we build a Bayesian data assimilation model that considers the full range of parameter trade-offs in estimation (Bauer et al., 2025b). We find that, in contrast to our conceptual

model, quickly ramped up SAI programs are only marginally more identifiable than linearly or slowly ramped up programs by mid-century. This is because of trade-offs in estimating the SAI versus the CO₂ climate response that cannot be disentangled on short timescales. We show that, counter-intuitively, the two effects with the largest impact on learning about the angle parameter are the magnitude of the background CO₂ forcing and the covariance structure between the SAI and CO₂ forcings. We close with a discussion of how these results could impact coupled CO₂ mitigation and SAI policies, as well as SAI scenario design.

2 Methods

2.1 Reduced Complexity Regional Climate Model

To emulate the global climate, we utilize a two-layer energy balance model (EBM) that is commonly used in reduced complexity climate emulators (Held et al., 2010; Geoffroy et al., 2013). For a specified pathway of CO₂ concentrations, C_t , and SAI forcing, f_t^{SAI} , we can write

$$c_{SL} \frac{dT_t^{SL}}{dt} = m \ln \left(\frac{C_t}{C_0} \right) - f_t^{SAI} - \lambda T_t^{SL} + \gamma \varepsilon (T_t^{DO} - T_t^{SL}) + q_t^{IV}, \quad (2.1)$$

$$c_{DO} \frac{dT_t^{DO}}{dt} = \gamma (T_t^{SL} - T_t^{DO}), \quad (2.2)$$

where T_t^{SL} is the surface layer temperature, T_t^{DO} is the deep ocean temperature, c_{SL} and c_{DO} are the heat capacities of each layer, m is the CO₂ forcing sensitivity, $C_0 = 278.3$ ppm is preindustrial CO₂ concentrations, $\lambda > 0$ is the climate feedback parameter, $\gamma > 0$ is the heat transfer coefficient, $\varepsilon > 0$ is the heat uptake efficiency, and q_t^{IV} is stochastic internal climate variability that is assumed to follow an AR(1) process.

We use pattern scaling to map global surface temperature T_t^{SL} to regional temperature $T_t^{(r)}$ for a set of regions $r \in R$ (Mitchell, 2003; Tebaldi and Arblaster, 2014). We alter the standard pattern scaling relationship with a correction term that includes how SAI changes the CO₂-only response, such that

$$T_t^{(r)} = \alpha_r T_t^{SL} + \beta_r f_t^{SAI} + q_t^{(r)}, \quad (2.3)$$

where α_r is the usual pattern scaling parameter, β_r is the SAI correction term, and $q_t^{(r)}$ is white regional noise. Note that $\beta_r \neq 0$ is required for $\vartheta \neq 0^\circ$. In all of our experiments, we choose a simple two region configuration; in the *Supplementary Information*, we show that generalizing to three regions does not qualitatively impact our results.

If we assume the deep ocean temperature adjusts slowly to forcing, we can utilize (2.1), (2.2), and (2.3) and the definition of the SAI angle parameter to write

$$\cos \vartheta = \frac{\boldsymbol{\alpha} \cdot \mathbf{d}}{\|\boldsymbol{\alpha}\| \|\mathbf{d}\|} \quad (2.4)$$

where we have used boldface symbols to denote vectors of parameters in each region, e.g., $\boldsymbol{\alpha} =$

114 $(\alpha_1, \alpha_2, \dots, \alpha_{|R|})^T$, defined

$$\mathbf{d} := (\lambda + \varepsilon\gamma)\boldsymbol{\beta} - \boldsymbol{\alpha} \quad (2.5)$$

115 for convenience, and $\|\boldsymbol{\alpha}\|$ ($\|\mathbf{d}\|$, resp.) is the length of the vector $\boldsymbol{\alpha}$ ($\mathbf{d}$, resp.). $\mathbf{d}$ quantifies the regional
 116 SAI cooling pattern accounting for deviations from the CO₂ warming pattern induced by $\boldsymbol{\beta} \neq 0$. For
 117 a full derivation, see the *Supplementary Information*.

118 Note that throughout we focus on the angle parameter for temperature; though one could formulate an
 119 angle parameter for precipitation, precipitation is noisier and SAI only indirectly impacts precipitation
 120 through changes in surface thermodynamics and atmospheric circulation (Dong et al., 2026). Therefore,
 121 the results we show in this paper can be thought of as a likely upper bound on learning for other aspects
 122 of the climate system related to SAI.

123 2.2 Bayesian Learning Model

124 2.2.1 Experimental Design

125 To compute optimal estimates of our model parameters conditional on pseudo-observations of the
 126 future, we utilize an ensemble of data assimilations (EDA; Isaksen et al., 2010). See the *Supplementary*
 127 *Information* for the technical details of our algorithm. EDA allows us to capture nonlinear transitions
 128 in posteriors that other approaches, such as a Kalman filter, cannot capture (Evensen et al., 2022).

129 Our setup treats each parameter in our regional reduced complexity climate emulator, $(\lambda, \gamma, \varepsilon, m, c_{SL}, c_{DO}, \boldsymbol{\alpha}, \boldsymbol{\beta})$,
 130 as uncertain. This is a substantive improvement over past work aimed at learning aspects of the cli-
 131 mate system from future pseudo-observations that treats only a few parameters as uncertain (Kelly
 132 and Kolstad, 1999; Leach, 2007; Padilla et al., 2011; J. Ring and E. Schlesinger, 2012; Urban et al.,
 133 2014; Kelly and Tan, 2015; Mori and Shiogama, 2018; Kelly et al., 2024), and allows us to capture the
 134 full set of parameter trade-offs in estimation.

135 In each of our numerical experiments, we choose model parameters, including the value of the SAI
 136 angle parameter, to be the “true”, data-generating values that we try to infer. Let $\vartheta^\dagger$ be the true value
 137 of the angle parameter. We vary $\vartheta^\dagger$ from 5° to 30° to probe how the underlying true value changes
 138 learning dynamics.

139 To probe how the level of atmospheric CO₂ concentrations might impact how quickly we can discern the
 140 SAI angle parameter, we utilize the high (H), medium (M), medium-overshoot (MO), low (L), very-low
 141 (VL), and very-low-high-overshoot (VLHO) CO₂ emissions scenarios from scenarioMIP7 (Van Vuuren
 142 et al., 2026) and utilize FaIR (Smith et al., 2024) to convert emissions to median CO₂ concentrations.

143 We set the amount of SAI forcing by computing the amount of forcing needed to achieve $\Delta T = 0.5^\circ\text{C}$
 144 or 1°C of global cooling when EBM parameters are set to their central values. The SAI forcing is
 145 ramped up from zero to its final value over the course of N years and held constant thereafter, implying
 146 that the global temperature change from SAI is given by

$$\Delta T_t^{\text{SAI}} = \begin{cases} -\Delta T(t/N)^\xi & 0 < t \leq N \\ -\Delta T & t > N \end{cases} \quad (2.6)$$

where ξ determines the “ramp rate” of the SAI program: $\xi < 1$ front-loads the SAI forcing ramp while $\xi > 1$ back-loads it. In our simulations, we use $\xi = 1/3$ as our fast ramp, $\xi = 1$ as our linear ramp, and $\xi = 3$ as our slow ramp. We stress that these choices are not meant to be an argument in favor of such a program; rather, they are meant to be representative of a moderate SAI program, in which context learning about the inequality of response would be relevant to decision makers (as opposed to, say, a unilateral deployment, where global inequality is less likely to factor into decision making).

Our experiments compute the optimal model parameters for 1000 ensemble members. We assimilate new observations of surface layer temperature, T_t^{SL} , ocean heat content, $Q_t := c_{SL}T_t^{SL} + c_{DO}T_t^{DO}$, and regional temperature $T_t^{(r)}$ every 5 years from 2025 until 2040, every 10 years from 2040 to 2060, again in 2075, and lastly in 2100, allowing us to probe rapid near-term learning as well as long-term uncertainty reductions efficiently. We warm start our model from 1850 to 2025 using only CO₂ concentrations to set the initial conditions for T_t^{SL} , T_t^{DO} and $T_t^{(r)}$ for $r \in R$.

2.2.2 Prior Calibration

We use the distribution of [Smith et al. \(2020\)](#) ($F_{2 \times CO_2} = 3.99 \pm 0.19 \text{ W m}^{-2}$) to set the prior for the CO₂ forcing sensitivity, m . The Intergovernmental Panel on Climate Change’s (IPCC) most recently estimated distribution for ECS (best estimate 3 °C with likely range 2.5-4 °C) then sets $\lambda = 1.3 \pm 0.31 \text{ W m}^{-2} \text{ K}^{-1}$ ([Sherwood et al., 2020](#); [Intergovernmental Panel On Climate Change \(IPCC\), 2023](#)). This prior for λ and the prior for ε taken from [Cummins et al. \(2020\)](#) set the prior for $\gamma = 0.62 \pm 0.26 \text{ W m}^{-2} \text{ K}^{-1}$ to match the best estimate distribution for TCR from the IPCC (best estimate 1.8 °C, likely range 1.4-2.2 °C; [Intergovernmental Panel On Climate Change \(IPCC\), 2023](#)). We lift the remaining global climate response parameter priors from [Leach et al. \(2021\)](#).

For the regional climate response parameters α and β , we use stylized priors that reproduce the distribution for ϑ used in [Harding et al. \(2023\)](#). The distribution used in [Harding et al. \(2023\)](#) was derived by synthesizing GCM output of SAI scenarios, from which they computed the angle parameter. This results in a median value of 15° and a 5-95 percentile range of 5°–30°. See the *Supplementary Information* for several tables detailing our prior calibration.

3 Results

3.1 Conceptual Model

To build intuition, we start with a simplified version of our Bayesian model. First, assume that the surface ocean layer is in quasi-equilibrium with the deep ocean, implying that the surface layer temperature T_t^{SL} will evolve linearly in forcing. This relationship was explored in [Bala et al. \(2010\)](#) and [Moreno-Cruz et al. \(2012\)](#) for SAI, and more recently for the global CO₂ response by [Jeevanjee et al. \(2026\)](#); each found that the relationship holds remarkably well on decadal timescales. These assumptions allow us to write the regional temperature as

$$T_t^{(r)} = \Gamma_{CO_2}^r \ln(C_t/C_0) + \Gamma_{SAI}^r \left[\Delta T(t/N)^\xi \right] + q_t^{(r)}, \quad (3.1)$$

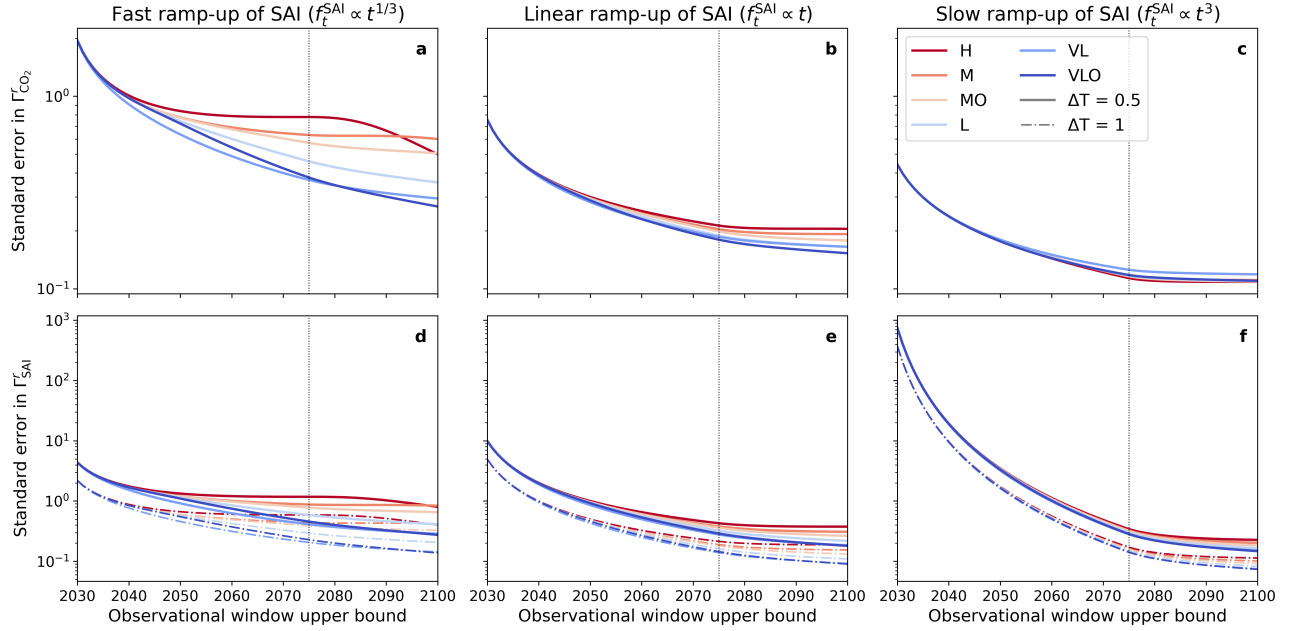


Figure 2: Conceptual Model Results. **a** shows the standard error in the estimation of $\Gamma_{\text{CO}_2}^r$, the regression coefficient of the CO_2 forcing component of regional temperature, as more observations are added to the estimation window when SAI is ramped up quickly. The color of each line represents a different CO_2 concentrations scenario (*Methods*). Solid lines represent a scenario where there is a total cooling by SAI of 0.5 °C, while dash-dot lines are scenarios for 1 °C total SAI cooling. The vertical dashed line is where SAI forcing is leveled off and held constant to maintain the targeted amount of cooling. **b** and **c** are as **a**, but when SAI is ramped up linearly and slowly, respectively. **d** shows the standard error in the estimation of Γ_{SAI}^r , the regression coefficient of the SAI-forced component of regional temperature, as more observations are assimilated when SAI is ramped quickly. **e** and **f** are as **d** but when SAI is ramped up linearly and slowly, respectively. Note SAI forcing is assumed to begin in 2025.

181 where $\Gamma_{\text{CO}_2}^r := \alpha_r m / \ell_{\text{eff}}$, $\Gamma_{\text{SAI}}^r := \beta_r \ell_{\text{eff}} - \alpha_r$, and $\ell_{\text{eff}} := \lambda + \gamma \varepsilon$ is the effective feedback. See the
 182 *Supplementary Information* for a full derivation. Interpreting (3.1), $\Gamma_{\text{CO}_2}^r$ is the regional temperature
 183 change per unit of logarithmic CO_2 forcing while Γ_{SAI}^r is regional temperature change per unit global
 184 SAI cooling. Note we have neglected global climate variability q_t^{IV} ; in the *Supplementary Information*
 185 we show that including it does not alter our results.

186 We assume that there is uncertainty in both $\Gamma_{\text{CO}_2}^r$ and Γ_{SAI}^r : we do not possess perfect knowledge to
 187 constrain the regional response to changes in global temperature, nor can we perfectly constrain how
 188 the regional response is different once we begin SAI. Uncertainties in the regional climate response
 189 to forcing have been growing in visibility recently, with climate models being unable to reproduce the
 190 observational record along a number of key axes (Shaw and Stevens, 2025). Therefore, these parameters
 191 must be jointly estimated throughout the SAI program.

192 We can use (3.1) to identify the controls on learning about the regional response to SAI. Suppose that
 193 an SAI program begins at $t_0 = 2025$ and is ramped up over 50 years, setting $N = 50$ in (3.1), and held
 194 constant thereafter. Assume we have n years of noisy observations of $T_t^{(r)}$. Since $q_t^{(r)}$ is assumed white,
 195 we can optimally estimate $\Gamma_{\text{CO}_2}^r$ and Γ_{SAI}^r and their standard errors (SEs) using multi-variate ordinary

least squares regression for different SAI cooling amounts, ramp rates, and CO₂ scenarios (*Methods*). The rate at which the SEs of $\Gamma_{\text{CO}_2}^r$ and Γ_{SAI}^r decline as we increase n gives us a sense of how quickly they could be learned during an SAI deployment.

The results of this calculation are shown in Figure 2. We find that learning about the CO₂ response is stratified by SAI scenarios, with quickly ramping up SAI scenarios (Fig. 2a) having larger SE than the linear or slow ramps (Figs. 2b,c, resp.). This is because the CO₂ forcing signal is confounded by the higher initial SAI forcing, leading to less learning about the CO₂-forced response. In each case, the SE falls by just under an order of magnitude by 2100. Learning about $\Gamma_{\text{CO}_2}^r$ is insensitive to the amount of SAI, as the amount of SAI does not interact with that component of the regression, see (3.1).

For the SAI response, we find that early (pre-2050) estimates of SE are smallest for fast SAI ramps, higher for linear ramps, and higher still for slow ramps. We further find that scenarios with more CO₂ forcing have less learning than those with lower CO₂ concentrations. These two effects can both be explained by the relative sizes of the SAI and CO₂ forcings on surface temperature.

What is surprising, however, is that the SE for Γ_{SAI}^r in the fast SAI ramp scenario is near-flat for the entire 21st century (Fig. 2d), with the SE being reduced only by about an order of magnitude by 2100. The linear and slow ramps, by comparison, have smaller SEs in 2100 than the fast ramp and more steeply declining learning curves (Figs. 2e-f). This can be explained by the covariance between logarithmic forcing and power law forcing that degrades learning: both the logarithm and power law are concave when $\xi < 1$. Therefore, when SAI forcing follows a fast ramp, the initial burst of signal lowers estimation error, but additional learning is hampered by covariance with the CO₂ forcing. The linear and slow ramps have higher initial uncertainties because of the initially small SAI signal, but the estimate of Γ_{SAI}^r is improved faster in these scenarios than the fast ramp because linear and convex forcings are more easily differentiated from the concave logarithmic CO₂ forcing. In both the linear and slow ramp cases, the gains to learning from anti-covariance more than make up for the initially smaller signals, leading to less error in Γ_{SAI}^r by 2100 compared to the fast ramp.

In the *Supplementary Information*, we provide two additional calculations to support this conclusion. The first is that, in the limit of infinitely many observations, the power ξ that maximizes error in the estimation of Γ_{SAI}^r is $\xi = 0$. (The logarithm is asymptotically flat.) The second calculation is that if CO₂ forcing were to be described itself by a power law, $f_t^{\text{CO}_2} \propto t^\varpi$, then $\varpi = \xi$ results in infinite estimation error. Furthermore, we show that changing the start date of the SAI program to 2075 does not materially change our findings. We also show that modulation of the SAI program (i.e., adding a small oscillatory component to the global SAI forcing) is moderately effective at speeding up learning of the regional response. It is worth noting that modulation may be more effective at speeding up learning of fast-responding parts of the climate system (like stratospheric circulation or chemistry). The size of this learning, as well as the social and political acceptability of such modulation, is an open question.

3.2 Bayesian Learning Model

We utilize our Bayesian learning model (*Methods*) to test if the insights from our conceptual model hold in a more realistic environment, where the global climate has auto-correlated internal variability

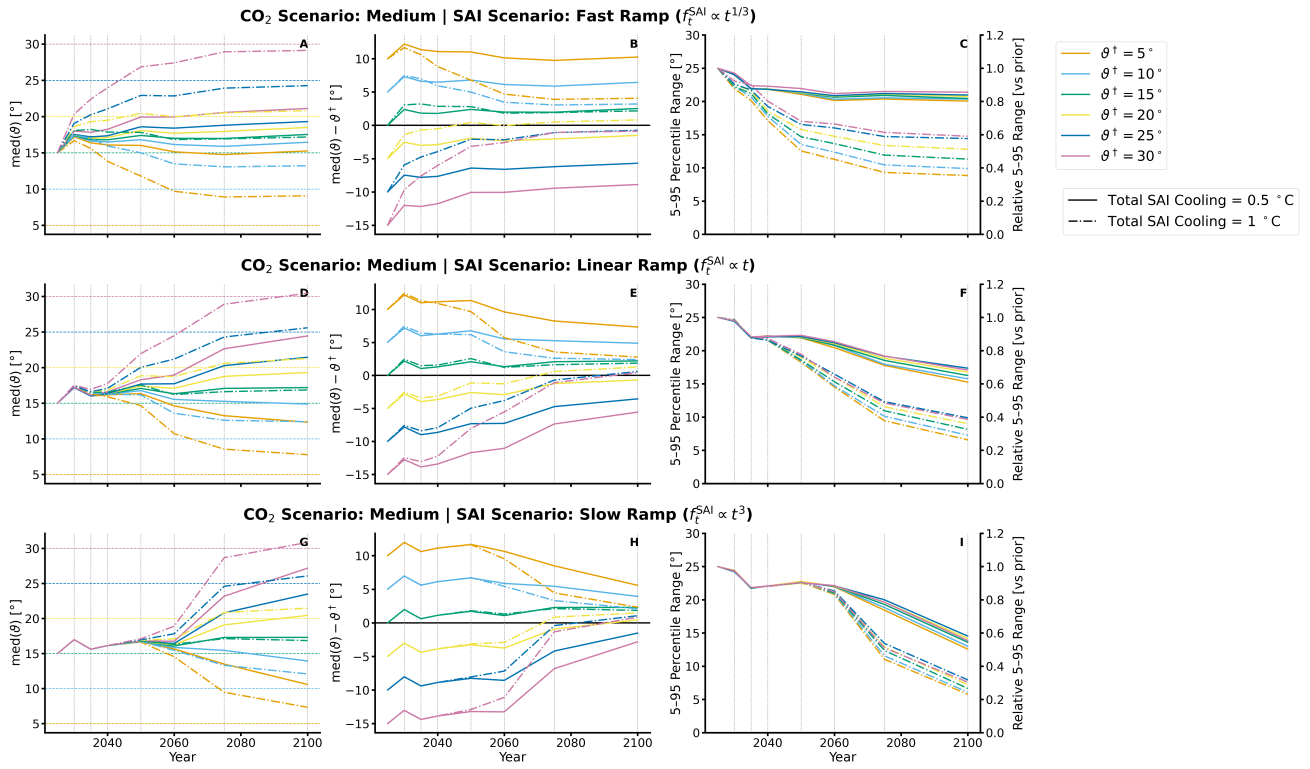


Figure 3: SAI Angle Learning Across SAI Scenario Designs. **A** shows the ensemble median estimate of the SAI angle parameter ϑ as a function of time for the Medium CO_2 and fast SAI ramp scenario (see the row title). Dotted vertical lines indicate times when new observations are assimilated into the posteriors. The colors represent different true values of the angle parameter. Solid lines (dash-dot, resp.) indicate posterior medians when the total SAI cooling is $0.5\text{ }^\circ\text{C}$ ($1\text{ }^\circ\text{C}$, resp.). Dashed horizontal lines show the true value of the angle parameter for reference. **B** shows the bias, defined as the difference between the ensemble median estimate of the angle parameter and its true value, as a function of time. **C** shows the 5-95 percentile range in absolute terms (left axis) and relative to the prior 5-95 percentile range (right axis). **D–F** and **G–I** are as **A–C**, but for the linear SAI scenario and the slowly ramped up SAI scenario, respectively.

and its response to forcing (i.e., ECS and TCR) is not *a priori* known, and updating model parameters depends both on fitting to data and the size of deviations from the prior.

Figure 3 shows the results from the Bayesian model holding the CO_2 scenario fixed (Medium) while changing the SAI ramp rate (rows) and total SAI cooling (linestyles). We find that across SAI scenarios with $0.5\text{ }^\circ\text{C}$ total cooling, the posterior median of the angle parameter never converges to the true value, leaving a lingering bias throughout the 21st century (Figs. 3A,B,D,E,G,H). Increasing the amount of SAI cooling results in low bias in 2100 only for the slow ramp; the fast and linear SAI ramps have nontrivial bias in 2100, indicating how difficult estimating the angle parameter proves to be once the full range of uncertainties about the climate system are taken into account.

Figure 3 indicates that the intuition we developed using our conceptual model holds for our Bayesian model, but the quantitative impact of each channel related to the SAI program (the size of the signal and the temporal covariance between the CO_2 and SAI forcings) are different. For bias in the angle parameter estimate (middle column of Figure 3), we find that the initial burst of signal provided by the fast SAI ramp only marginally lowers near-term bias in the $1\text{ }^\circ\text{C}$ cooling case; near-term bias in

the 0.5 °C case is about the same across SAI scenarios. However, end-of-century bias is substantially lower across total SAI cooling amounts for the slow ramp relative to the linear and fast ramps, as anti-covariance between the CO₂ and SAI forcings boosts learning later in the program.

These results also hold for uncertainty in the angle posterior (right column of Figure 3). Initial levels of uncertainty are roughly constant in the first 10-15 years of the program in each SAI scenario; the only exception is the 1 °C cooling fast ramp (Fig. 3C) which has small early uncertainty reductions owing to a higher initial SAI signal. For the fast SAI ramp, uncertainty is about constant after 2050, as learning is limited by the temporal covariance between the CO₂ and SAI forcings. We find that uncertainty in 2100 is lowest for the slow ramp, followed by the linear ramp, and the fast ramp having the highest uncertainty in 2100.

Taken together, these results suggest that the covariance effect has a larger quantitative impact on learning than the SAI signal size effect. Indeed, comparing 2100 levels of uncertainty in the angle parameter suggests that a slowly ramped up SAI program with 0.5 °C cooling has comparable uncertainty to a quickly ramped up program with 1 °C cooling.

Figure 4 quantifies the effect of the CO₂ forcing signal size on learning about the angle parameter. We find that learning, both in terms of lowering bias in the estimate of the angle parameter (middle column of Figure 4) and uncertainty (right column of Figure 4), is modulated by the level of CO₂ concentrations, corroborating the intuition developed in our conceptual model.

Our model suggests that the quantitative impact of the CO₂ forcing channel of learning can be substantial. For example, end-of-century uncertainty in the High emissions scenario is larger than in the Low emissions scenario by about 40%. Bias in the estimate of the angle parameter is reduced at a more rapid rate when CO₂ emissions are lower (see Figs. 4H,K) than when they are elevated for longer (Fig. 4B). This indicates a counter-intuitive channel through which learning about the inequality of the SAI response can be accelerated: by limiting CO₂ emissions throughout the program.

4 Implications for SAI and CO₂ Mitigation Policy

Our results have two high-level implications for SAI and CO₂ mitigation policy, and how they might be coupled. While our conceptual model (Section 3.1) identified three mechanisms that can influence learning about the regional inequality of SAI – the size of the SAI forcing, the size of the CO₂ forcing, and the temporal correlation between the two forcings – our quantitative results (Section 3.2, Figs. 3 and 4) suggest that the latter two effects dominate the first. This is because the burst of SAI signal from quickly ramping up the program is too short-lived to be disentangled from uncertainties in the global CO₂ response; on the other hand, limiting CO₂ concentrations dampens the confounding CO₂ signal and enables more learning about the SAI response. Further exploiting the correlation structure of the forcings in time only improves detectability.

These findings provide an argument for coupling commitments to cut, or at least hold constant, CO₂ emissions with a potential SAI program based on SAI inequality risk management. Our finding that more CO₂ emissions during an SAI program lead to slower uncertainty resolution in the angle parameter implies increased uncertainty in climate impacts during the deployment if CO₂ concentrations remain

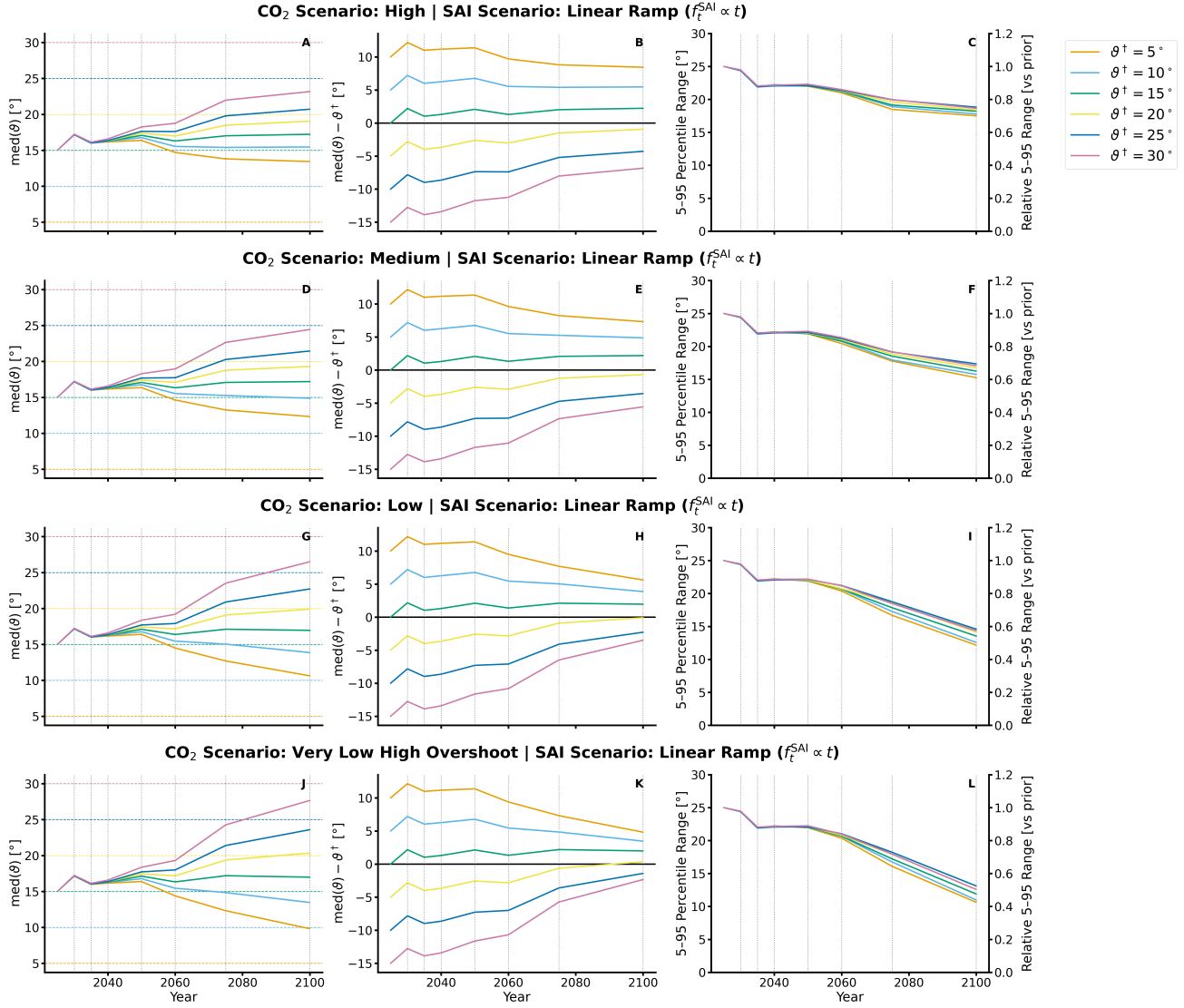


Figure 4: SAI Angle Learning Across CO₂ Scenarios. **A** shows the ensemble median estimate of the SAI angle parameter ϑ as a function of time for the High CO₂ and linear SAI ramp scenario (see the row title). Dotted vertical lines indicate times when new observations are assimilated into the posteriors. The colors represent different true values of the angle parameter. Dashed horizontal lines show the true value of the angle parameter for reference. **B** shows the bias, defined as the difference between the ensemble median estimate of the angle parameter and its true value, as a function of time. **C** shows the 5-95 percentile range in absolute terms (left axis) and relative to the prior 5-95 percentile range (right axis). **D–F**, **G–I**, and **J–L** are as **A–C**, but for the Medium, Low, and Very Low High Overshoot scenarios, respectively. Note total cooling from SAI is 0.5 °C.

287 elevated. This, in turn, implies a risk-averse and adaptive rational decision-maker (in the utility-
288 maximizing sense) would curb the level of mitigation deterrence – that is, the extent to which one
289 emits more CO₂ as a result of SAI deployment (Moreno-Cruz et al., 2025) – to lessen SAI inequality
290 risks. Of course, the ultimate determinant of decisions regarding SAI deployment will not be made
291 solely on the basis of physical impacts, nor should it be. However, this insight can be incorporated
292 into a more holistic view of SAI policy, including ethics and climate justice, in determining potential
293 couplings between CO₂ mitigation and SAI policy.

294 Finally, our results provide an additional benefit of slowly ramping up any potential SAI program: the
295 inequality in benefits of such a program will be more easily discernible relative to a hastily ramped
296 up program. This complements an assortment of reasons why a slowly ramped up program would be
297 prudent, including the ability to monitor any changes in stratospheric circulation and chemistry or
298 interactions between falling SAI aerosols and upper tropospheric aerosols (Tilmes et al., 2009; Diallo
299 et al., 2017; Bednarz et al., 2023). Future work could probe the needed perturbation sizes to constrain
300 these processes subject to observations during a potential deployment in a similar framework as we
301 have established here.

302 5 Conclusion

303 We have presented a hierarchy of models to understand how quickly we may be able to discern the
304 SAI angle parameter – a measurement of climate response inequality between a climate forced with
305 CO₂-only and CO₂ and SAI – during a potential deployment. We show that the size of the SAI forcing
306 signal, the size of the confounding CO₂ forcing signal, and the temporal covariance between these two
307 signals each play a role in the learning rate of the angle parameter. However, our model suggests that
308 the latter two effects are quantitatively more important than the first, implying that limiting CO₂
309 emissions and slowly ramping up SAI are the primary policy levers that can accelerate our ability to
310 detect inequalities in the SAI climate response during a potential deployment.

311 Acknowledgments

312 This work benefited from helpful comments from Juan Moreno-Cruz, Vikrant Sapkota, Rahul Singh,
313 and B. B. Cael’s research group. AMB was supported by the Institute for Climate and Sustainable
314 Growth and the Climate Systems Engineering Initiative at the University of Chicago. We are grateful
315 to have had access to the Research Computing Cluster at the University of Chicago that enabled us to
316 carry out the numerical work in this paper.

317 Data Availability Statement

318 All data presented in this study were generated using simulations. The codebase to carry out all analysis
319 in this paper can be found at https://github.com/adam-bauer-34/learning_sai. The codebase was
320 originally written by AMB and later refined using Claude. No artificial intelligence was used in the
321 conceptualization or writing of this manuscript. All data will be frozen in Zenodo upon acceptance.

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Supplementary Information for: “Inferring Stratospheric Aerosol Injection Climate Response Inequality”

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October 8, 2026

1 Full Derivation: SAI Angle Parameter

Consider our model outlined in the *Methods* section of the main text. Throughout this derivation, we will ignore stochastic terms in our equations, as the angle parameter is meant to be a noise-independent measure of climate response inequality. Allow the surface ocean to be in quasi-equilibrium: this implies that the deep ocean temperature, T_t^{DO} , is slowly adjusting to forcing, such that $dT_t^{DO}/dt \approx 0$ and $T_t^{DO} \approx 0$. Then only the surface layer adjusts to forcing, and its evolution is given by

$$c_{SL} \frac{dT_t^{SL}}{dt} = f_t^{CO_2} - f_t^{SAI} - (\lambda + \gamma\varepsilon)T_t^{SL}, \quad (1.1)$$

where we have defined $f_t^{CO_2} := m \ln(C_t/C_0)$. Now allow the surface layer to equilibrate on fast timescales (Held et al., 2010). This sets $dT_t^{SL}/dt \approx 0$ and allows us to write

$$T_t^{SL} \approx \frac{f_t^{CO_2} - f_t^{SAI}}{\lambda + \gamma\varepsilon}. \quad (1.2)$$

This model is equivalent to the “zero-layer” model of Gregory et al. (2015). Jeevanjee et al. (2026) found this model holds remarkably well on decadal timescales, though on millennial timescales the deep ocean component of the model was shown to be lacking nonlinear behavior found in climate models.

Using our pattern scaling relationship from the main text, we can now write the main two regional climate vectors needed in our calculation. Allow α and β to be pattern scaling parameter vectors

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over the set of regions R following the *Methods* section in the main text. Then first we can write the CO₂-only temperature response vector as

$$\mathbf{T}_t^{CO_2-only} = \boldsymbol{\alpha} \left(\frac{f_t^{CO_2}}{\lambda + \gamma\varepsilon} \right). \quad (1.3)$$

Second, we can write the regional temperature vector with CO₂ and SAI forcing such that

$$\begin{aligned} \mathbf{T}_t^{CO_2+SAI} &= \boldsymbol{\alpha} T_t^{AT} + \boldsymbol{\beta} f_t^{SAI} \\ &= \underbrace{\boldsymbol{\alpha} \left(\frac{f_t^{CO_2}}{\lambda + \gamma\varepsilon} \right)}_{\equiv \mathbf{T}_t^{CO_2-only}} + f_t^{SAI} \left(\boldsymbol{\beta} - \frac{\boldsymbol{\alpha}}{\lambda + \gamma\varepsilon} \right) \\ &= \mathbf{T}_t^{CO_2-only} + f_t^{SAI} (\lambda + \gamma\varepsilon) \mathbf{d} \end{aligned} \quad (1.4)$$

where we have defined

$$\mathbf{d} := (\lambda + \varepsilon\gamma)\boldsymbol{\beta} - \boldsymbol{\alpha} \quad (1.5)$$

for convenience. We can now formulate the residual vector shown in Figure 1 in the main text, $\mathbf{R}_t$, as

$$\mathbf{R}_t = \mathbf{T}_t^{CO_2+SAI} - \mathbf{T}_t^{CO_2-only} = f_t^{SAI} (\lambda + \gamma\varepsilon) \mathbf{d}. \quad (1.6)$$

To complete the derivation, we can utilize the definition of the dot product between two vectors to write

$$\begin{aligned} \mathbf{T}_t^{CO_2-only} \cdot \mathbf{R}_t &= \left(\frac{f_t^{CO_2}}{\lambda + \gamma\varepsilon} \right) (f_t^{SAI} (\lambda + \gamma\varepsilon)) \boldsymbol{\alpha} \cdot \mathbf{d} = \left\| \mathbf{T}_t^{CO_2-only} \right\| \left\| \mathbf{R}_t \right\| \cos \vartheta \\ &= \left(\frac{f_t^{CO_2}}{\lambda + \gamma\varepsilon} \right) (f_t^{SAI} (\lambda + \gamma\varepsilon)) \|\boldsymbol{\alpha}\| \|\mathbf{d}\| \cos \vartheta \end{aligned}$$

which simplifies to

$$\cos \vartheta = \frac{\boldsymbol{\alpha} \cdot \mathbf{d}}{\|\boldsymbol{\alpha}\| \|\mathbf{d}\|}. \quad (1.7)$$

2 Supporting Calculations: Conceptual Model

2.1 Full Derivation of Regression Model

Following (1.2) and adding back the stochastic component, we can write

$$T_t^{SL} \approx \frac{m \ln(c_t) - f_t^{SAI} + q_t^{IV}}{\ell_{eff}}, \quad (2.1)$$

where $\ell_{eff} := \lambda + \gamma\varepsilon$ and c_t is ratio of CO₂ concentrations to preindustrial. Let the forcing from SAI follow

$$f_t^{SAI} = [\Delta T \ell_{eff}] (t/N)^\xi \quad (2.2)$$

which translates (2.6) in the main text to a forcing using the linearity assumption above.

Using (2.1) in (2.3) in the main text and simplifying, we can write the regional temperature as

$$\begin{aligned} T_t^{(r)} &= \alpha_r \left(\frac{m \ln(c_t) - [\Delta T \ell_{eff}] (t/N)^\xi + q_t^{IV}}{\ell_{eff}} \right) + \beta_r [\Delta T \ell_{eff}] (t/N)^\xi + q_t^r, \\ &= \underbrace{\left(\frac{\alpha_r m}{\ell_{eff}} \right)}_{=: \Gamma_{CO_2}^r} \ln(c_t) + \underbrace{(\beta_r \ell_{eff} - \alpha_r)}_{=: \Gamma_{SAI}^r} \Delta T (t/N)^\xi + \underbrace{\left(\frac{\alpha_r}{\ell_{eff}} \right) q_t^{(r)}}_{\equiv \Gamma_{CO_2}^r / m} q_t^{(r)} \\ &= \Gamma_{CO_2}^r \ln(c_t) + \Gamma_{SAI}^r [\Delta T (t/N)^\xi] + \left(\frac{\Gamma_{CO_2}^r}{m} \right) q_t^{IV} + q_t^r. \end{aligned} \quad (2.3)$$

as desired. Eqn. (3.1) in the main text is (2.3) with q_t^{IV} set to zero. Figure S1 shows that the main conclusions of our conceptual model is unchanged when we estimate $(\Gamma_{CO_2}^r, \Gamma_{SAI}^r)$ including the influence of q_t^{IV} using generalized least squares.

2.2 Supporting Calculations

In the following calculations, we support our main findings of our conceptual model. Here, we neglect the plateau in SAI forcing and allow SAI forcing to follow a power law, $f_t^{SAI} = t^\xi$, as one can show that factors of ΔT and $N^{-\xi}$ do not impact the results below.

Proposition 2.1. *For our main regression model (3.1) in the main text, in the limit of infinitely many observations, the standard error in Γ_{SAI}^r is maximized when $\xi = 0$.*

Proof. Consider (3.1) in the main text and suppose there are n observations of $T_t^{(r)}$. Suppose that the long-run forcing from CO₂ equals $\mathcal{L} > 0$. Let $\vec{\pi} := (\Gamma_{CO_2}^r, \Gamma_{SAI}^r)^T$ be the vector of regression parameters and

$$\mathbf{X} := \begin{pmatrix} \ln c_1 & \ln c_2 & \cdots & \ln c_n \\ 1 & 2^\xi & \cdots & n^\xi \end{pmatrix} \quad (2.4)$$

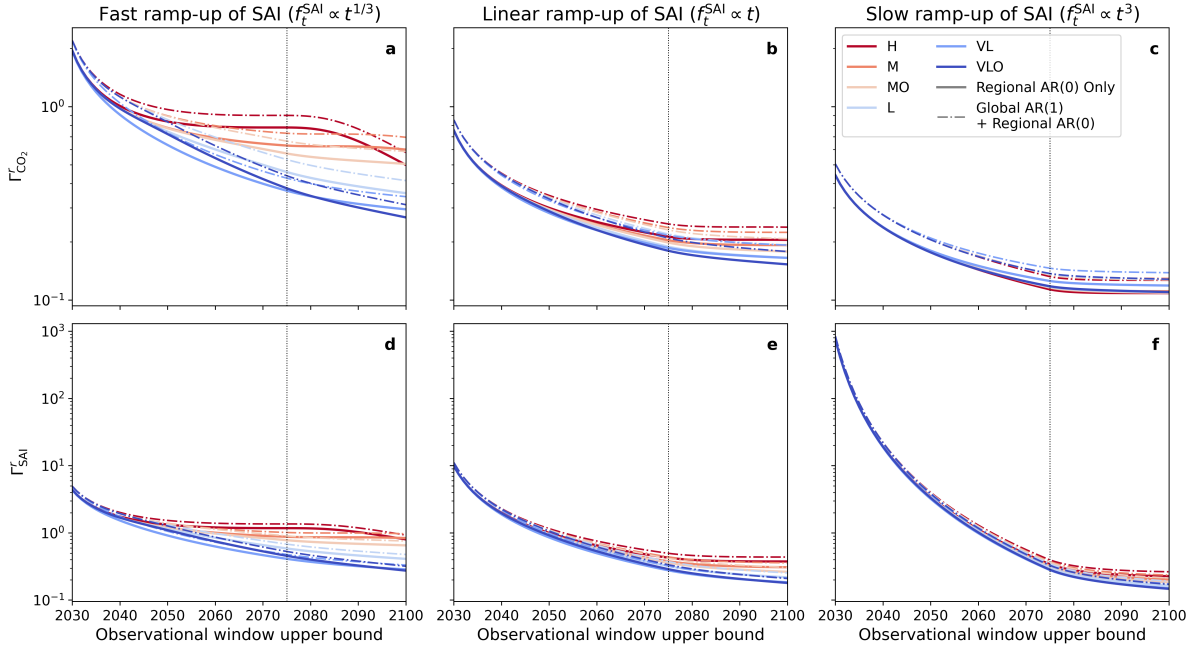


Figure S1: Conceptual Model Results with Global Noise. **a** shows the standard error in the estimation of $\Gamma_{\text{CO}_2}^r$, the regression coefficient of the CO_2 forcing component of regional temperature, as more observations are added to the assimilation window when SAI is ramped up quickly. The color of each line represents a different CO_2 concentrations scenario (*Methods*). Solid lines represent a scenario with only AR(0) regional noise, while dash-dot lines include the influence of global AR(1) noise. The vertical dashed line is where SAI forcing is leveled off and held constant to maintain targeted amount of cooling. **b** and **c** are as **a**, but when SAI is ramped up linearly and slowly, respectively. **d** shows the standard error in the estimation of Γ_{SAI}^r , the regression coefficient of the SAI-forced component of regional temperature, as more observations are assimilated when SAI is ramped quickly. **e** and **f** are as **d** but when SAI is ramped up linearly and slowly, respectively. Note SAI forcing is assumed to begin in 2025.

be the design matrix. For ordinary least squares regression (OLS), the covariance matrix for the estimate of $\vec{\pi}$, $\text{Cov}(\vec{\pi})$, is proportional to

$$\begin{aligned}\text{Cov}(\vec{\pi}) &= \sigma^{-2} (\mathbf{X}^T \mathbf{X})^{-1} \\ &= \frac{\sigma^{-2}}{AC(\xi) - B(\xi)^2} \begin{pmatrix} C(\xi) & -B(\xi) \\ -B(\xi) & A \end{pmatrix}\end{aligned}\quad (2.5)$$

where

$$A := \sum_{t=1}^n \ln^2 c_t, \quad B(\xi) := \sum_{t=1}^n t^\xi \ln c_t, \quad C(\xi) := \sum_{t=1}^n t^{2\xi} \quad (2.6)$$

and σ^2 is the variance of $q_t^{(r)}$. Choosing the lower right diagonal element of (2.5) results in the standard error (SE) of Γ_{SAI}^r as

$$\text{SE}(\Gamma_{\text{SAI}}^r)^2 \propto \frac{A}{AC(\xi) - B(\xi)^2}. \quad (2.7)$$

As our goal is to find the ξ that maximizes the SE of Γ_{SAI}^r , we can differentiate both sides of (2.7) and simplify to find

$$\frac{\partial (\ln \text{SE}(\Gamma_{\text{SAI}}^r)^2)}{\partial \xi} = - \left[\frac{1}{AC(\xi) - B(\xi)^2} \right] \left[A \frac{\partial C(\xi)}{\partial \xi} - 2B(\xi) \frac{\partial B(\xi)}{\partial \xi} \right]. \quad (2.8)$$

By Lagrange's identity, $AC(\xi) - B(\xi)^2 > 0$ for all $\xi > 0$.

The strategy for the remainder of the proof is to determine the sign of $\partial(\ln \text{SE}(\Gamma_{\text{SAI}}^r)^2)/\partial \xi$. If it can be shown that $\partial(\ln \text{SE}(\Gamma_{\text{SAI}}^r)^2)/\partial \xi < 0$ for all $\xi > 0$ in the limit of large n , it immediately follows that $\xi = 0$ maximizes $\text{SE}(\Gamma_{\text{SAI}}^r)$. For ease of notation, define

$$a_t := \ln c_t, \quad b_t = t^\xi, \quad c_t = \ln t, \quad (2.9)$$

and write

$$\Xi(\xi) := A \frac{\partial C(\xi)}{\partial \xi} - 2B(\xi) \frac{\partial B(\xi)}{\partial \xi} = \sum_{t=1}^n a_t^2 \sum_{t=1}^n b_t^2 c_t - \sum_{t=1}^n b_t a_t \sum_{t=1}^n b_t a_t c_t \quad (2.10)$$

which can be written as

$$\Xi(\xi) = 2 \sum_{t=1}^n \sum_{\nu=1}^n c_\nu [(a_t^2 b_\nu^2 - a_t b_t a_\nu b_\nu)] \quad (2.11)$$

where we have reindexed the second sum in (2.10) using a dummy index. Carrying out the same procedure for the denominator of (2.8) results in

$$\Upsilon(\xi) := AC(\xi) - B(\xi)^2 = \sum_{t=1}^n \sum_{\nu=1}^n [a_t^2 b_\nu^2 - a_t b_t a_\nu b_\nu] \quad (2.12)$$

Using (2.11) and (2.12) in (2.8) results in

$$\frac{\partial (\ln \text{SE}(\Gamma_{\text{SAI}}^r))}{\partial \xi} = -2 \frac{\sum_{t=1}^n \sum_{\nu=1}^n c_\nu [(a_t^2 b_\nu^2 - a_t b_t a_\nu b_\nu)]}{\sum_{t=1}^n \sum_{\nu=1}^n [a_t^2 b_\nu^2 - a_t b_t a_\nu b_\nu]}. \quad (2.13)$$

In the limit of large n , we can approximate the sums in (2.13) with integrals from $0 \rightarrow n$. Then (2.12) becomes

$$\Upsilon(\xi) \approx \mathcal{L}^2 n^{2\xi+2} \left(\frac{\xi^2}{(2\xi+1)(\xi+1)^2} \right) \quad (2.14)$$

and the numerator is

$$\Xi(\xi) \approx \mathcal{L}^2 n^{2\xi+2} \left(\frac{\xi^2 \ln n}{(2\xi+1)(\xi+1)^2} - \frac{\xi^2 - \xi - 1}{(2\xi+1)^2(\xi+1)^3} \right). \quad (2.15)$$

Using (2.15) and (2.14) in (2.13) results in

$$\frac{\partial (\ln \text{SE}(\Gamma_{\text{SAI}}^r))}{\partial \xi} \approx -2 \ln n + \frac{\xi^2 - \xi - 1}{\xi(\xi+1)(2\xi+1)}. \quad (2.16)$$

The sign of $\partial(\ln \text{SE}(\Gamma_{\text{SAI}}^r))/\partial \xi$ is therefore determined by the relative sizes of $-\ln n$ and $(\xi^2 - \xi - 1)/(\xi(2\xi+1)(\xi+1))$. One can show numerically that

$$\max_{\xi} \left(\frac{\xi^2 - \xi - 1}{\xi(\xi+1)(2\xi+1)} \right) \approx 0.0615663 \quad (2.17)$$

which occurs when $\xi = \xi^* \approx 3.62045$. Therefore,

$$\frac{\partial (\ln \text{SE}(\Gamma_{\text{SAI}}^r))}{\partial \xi} < 0 \iff n > e^{0.0615663} \approx 1.1 \quad (2.18)$$

which is trivially satisfied as we assumed n is large. Therefore, for all $\xi > 0$,

$$\frac{\partial (\ln \text{SE}(\Gamma_{\text{SAI}}^r))}{\partial \xi} < 0 \quad (2.19)$$

when n is large, meaning $\xi = 0$ maximizes $\text{SE}(\Gamma_{\text{SAI}}^r)$, completing the proof. $\square$

Proposition 2.2. For our main regression model (3.1) in the main text, if CO_2 forcing follows a power law relationship in time such that $f_t^{\text{CO}_2} = mt^\varpi$, $\varpi = \xi$ leads to infinite error in Γ_{SAI}^r .

Proof. Let there be $n > 2$ observations of $T_t^{(r)}$. Consider (3.1) in the main text and replace the $\ln(c_t)$ with t^ϖ , giving

$$T_t^{(r)} = \Gamma_{\text{CO}_2}^r t^\varpi + \Gamma_{\text{SAI}}^r t^\xi + q_t^{(r)}, \quad (2.20)$$

73 which has a design matrix of

$$\mathbf{X} = \begin{pmatrix} 1 & 2^\varpi & \dots & n^\varpi \\ 1 & 2^\xi & \dots & n^\xi \end{pmatrix}. \quad (2.21)$$

74 Using OLS, we can write the covariance matrix of $\vec{\pi} := (\Gamma_{\text{CO}_2}^r, \Gamma_{\text{SAI}}^r)^T$ as

$$\begin{aligned} \text{Cov}(\vec{\pi}) &= \sigma^{-2} (\mathbf{X}^T \mathbf{X})^{-1} \\ &= \frac{\sigma^{-2}}{CP - G^2} \begin{pmatrix} P & -G \\ -G & C \end{pmatrix} \end{aligned} \quad (2.22)$$

75 where we have defined

$$C := \sum_{t=1}^n t^{2\varpi}, \quad (2.23)$$

$$G := \sum_{t=1}^n t^{\varpi+\xi}, \quad (2.24)$$

$$P := \sum_{t=1}^n t^{2\xi}. \quad (2.25)$$

76 Focusing on the prefactor of (2.22), we can write

$$\text{Cov}(\vec{\pi}) \propto \frac{1}{\sum_{t=1}^n \sum_{\nu=1}^n [t^{2\varpi} \nu^{2\xi} (1 - \nu^{\varpi-\xi} t^{\xi-\varpi})]}. \quad (2.26)$$

77 It immediately follows that $\varpi = \xi$ implies $\text{Cov}(\vec{\pi}) \rightarrow \infty$, as desired. Intuitively, one can see that
 78 if $\varpi = \xi$, a regression will only identify $\Gamma_{\text{CO}_2}^r + \Gamma_{\text{SAI}}^r$, rather than either term individually, making
 79 estimation of Γ_{SAI}^r infeasible. $\square$

80 2.3 Additional Figures

81 Figure S2 shows that our results in the main text do not change when SAI is started later in the
 82 century (here in 2075). Figure S3 shows the results of our conceptual model with global SAI forcing
 83 is given by

$$f_t^{\text{SAI}} = [\Delta T \ell_{\text{eff}}] (t/N)^\xi + 0.1 \ell_{\text{eff}} \cos(t - t_0). \quad (2.27)$$

84 Adding a small oscillatory component to the SAI forcing tends to improve near-term learning
 85 from the slowly and linearly ramped up SAI scenarios. We stress that the social and political
 86 acceptability of modulating the SAI forcing signal is an open question.

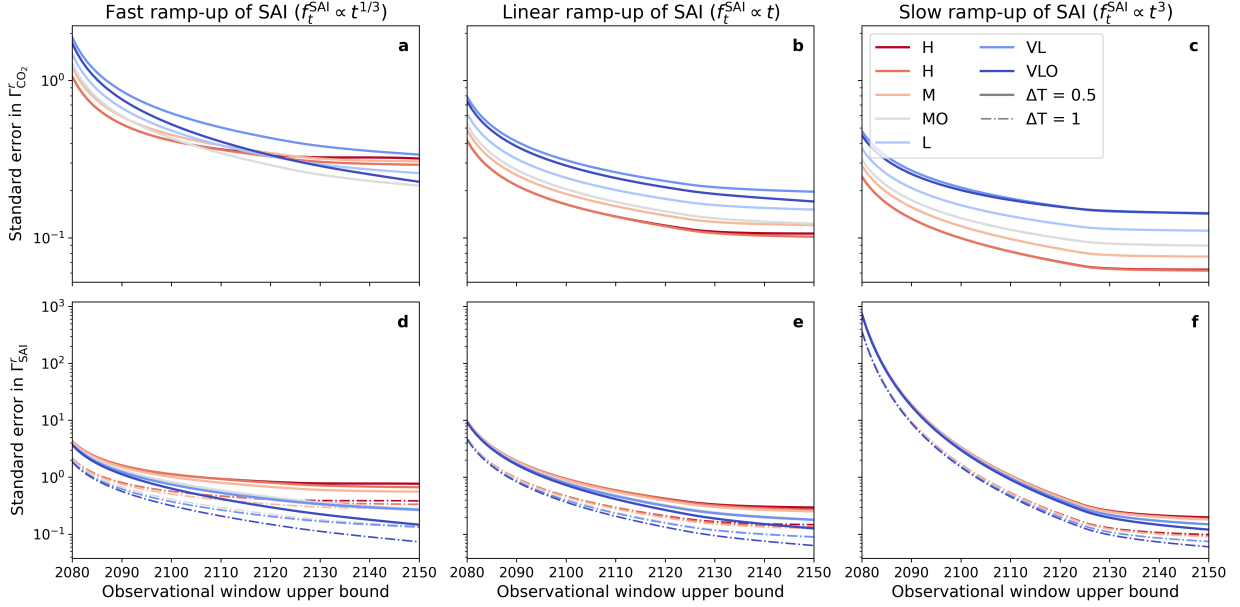


Figure S2: Conceptual Model where SAI begins in 2075. **a** shows the standard error in the estimation of $\Gamma_{\text{CO}_2}^r$, the regression coefficient of the CO_2 forcing component of regional temperature, as more observations are added to the assimilation window when SAI is ramped up quickly. The color of each line represents a different CO_2 concentrations scenario (*Methods*). Solid lines represent a scenario where there is a total cooling by SAI of 0.5°C , while dash-dot lines are scenarios for 1°C total SAI cooling. The vertical dashed line is where SAI forcing is leveled off and held constant to maintain targeted amount of cooling. **b** and **c** are as **a**, but when SAI is ramped up linearly and slowly, respectively. **d** shows the standard error in the estimation of Γ_{SAI}^r , the regression coefficient of the SAI-forced component of regional temperature, as more observations are assimilated when SAI is ramped quickly. **e** and **f** are as **d** but when SAI is ramped up linearly and slowly, respectively. Note SAI forcing is assumed to begin in 2075.

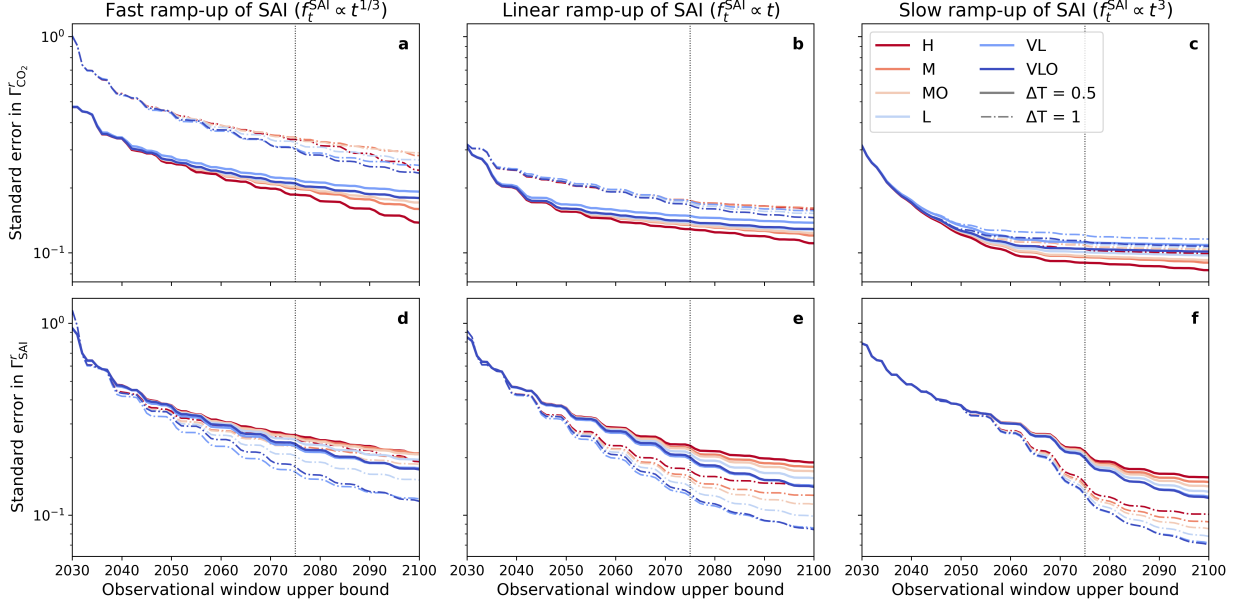


Figure S3: Conceptual Model with Oscillatory SAI Forcing. **a** shows the standard error in the estimation of $\Gamma_{\text{CO}_2}^r$, the regression coefficient of the CO_2 forcing component of regional temperature, as more observations are added to the assimilation window when SAI is ramped up quickly. The color of each line represents a different CO_2 concentrations scenario (*Methods*). Solid lines represent a scenario where there is a total cooling by SAI of 0.5°C , while dash-dot lines are scenarios for 1°C total SAI cooling. The vertical dashed line is where SAI forcing is leveled off and held constant to maintain targeted amount of cooling. **b** and **c** are as **a**, but when SAI is ramped up linearly and slowly, respectively. **d** shows the standard error in the estimation of Γ_{SAI}^r , the regression coefficient of the SAI-forced component of regional temperature, as more observations are assimilated when SAI is ramped quickly. **e** and **f** are as **d** but when SAI is ramped up linearly and slowly, respectively. Note SAI forcing is assumed to begin in 2025 and SAI global cooling is modulated at an annual frequency with amplitude 0.1°C .

3 Additional Details on Ensemble of Data Assimilations

3.1 Outline of Algorithm

In this section, we give a step-by-step description of our ensemble of data assimilations (EDA) algorithm we use to estimate model parameters conditional on pseudo-observations of the future climate.

1. **Initialization.** Choose:

- CO₂ concentrations scenario
- SAI °C total cooling
- The number of years to ramp up the SAI forcing (held constant after this number of years)
- The year to start the SAI program
- The year to start data assimilations, t_0
- A set of assimilation windows (t_0, t^*) for various t^* that define when observations are assimilated into the posterior
- The true (data-generating) climate model parameters including ϑ
- The number of ensemble members N_{ens}

2. **Warm start.** Simulate the reduced complexity climate model from 1850 to t_0 . Set the initial condition priors for state variables $(T_t^{SL}, T_t^{DO}, T_t^{(r)})$ using the results of the warm start.

3. **Data generation.** Use the true (data-generating) climate model parameters to simulate noisy pseudo-observations of the climate system from t_0 to 2100. Let this vector of observed quantities be given by $(\vec{T}_{SL,obs}, \vec{Q}_{obs}, \vec{T}_{r,obs})^T$ where the vectors are over time steps.

4. **Set priors.** Initialize priors for initial conditions, model parameters, and noise.

5. **Draw from priors.** Draw N_{ens} samples from the prior initialized in the previous step.

6. **Loop through assimilation windows** $(t_0, t^*) \forall t^*$:

- **Generate time path for each ensemble member.** Simulate the reduced complexity climate model for the n^{th} ensemble member parameter vector $\vec{x}^{(n)}$. This results in a set of data given by $(\vec{T}_{SL}^{(n)}, \vec{Q}^{(n)}, \vec{T}_r^{(n)})^T$.

- **Minimize the cost function for each ensemble member.** The cost function for the ensemble member n is given by

$$\begin{aligned}
J^n(\vec{x}^{(n)}) = & \left(\vec{x}^{(n)} - \vec{x}_0\right)^T \Sigma_0^{-1} \left(\vec{x}^{(n)} - \vec{x}_0\right) + \left(\vec{T}_{SL}^{(n)} - \vec{T}_{SL,obs}\right)^T \left(\vec{T}_{SL}^{(n)} - \vec{T}_{SL,obs}\right) \\
& + \left(\vec{Q}^{(n)} - \vec{Q}_{obs}\right)^T \left(\vec{Q}^{(n)} - \vec{Q}_{obs}\right) \\
& + \sum_{r \in R} \left(\vec{T}_r^{(n)} - \vec{T}_{r,obs}\right)^T \left(\vec{T}_r^{(n)} - \vec{T}_{r,obs}\right)
\end{aligned} \tag{3.1}$$

where $\vec{x}_0$ is the prior central estimate of the the state vector and Σ_0^{-1} is the inverse covariance matrix of the prior. This cost function can be derived by using Gaussian priors for misfits in the data and the observations (see [Evensen et al., 2022](#), for a full theoretical treatment). We utilize the adjoint method ([Errico, 1997](#); [Park and Zupanski, 2022](#)) to compute the cost function gradient in the minimization step to ensure convergence.

7. Collect and store results.

3.2 Verifying the Data Assimilation Model

To verify that our EDA algorithm has been coded correctly, we compute two diagnostics:

$$R_\alpha := \frac{\|\mathbf{M}(\vec{X}_0 + \alpha \delta \vec{X}_0) - \mathbf{M}(\vec{X}_0)\|}{\|\alpha \mathbf{L} \delta \vec{X}_0\|} \tag{3.2}$$

and

$$\Phi_\alpha := \frac{J(\vec{X} + \alpha \vec{h}) - J(\vec{X})}{\alpha \vec{h}^T \vec{\nabla} J(\vec{X})}. \tag{3.3}$$

The first represents the extent to which the tangent linear model, $\mathbf{L}$, appropriately approximates the nonlinear propagation operator $\mathbf{M}$ for our reduced complexity climate model for increasingly small perturbations $\alpha > 0$ around some state vector $\vec{X}_0$. The second represents the same computation but for the gradient of the cost function J , which is vital in our algorithm’s Bayesian update routine as it relies on the adjoint of the nonlinear operator $\mathbf{M} = \mathbf{L}^T$. See Figure S5: both diagnostic quantities follow the typical “V”-shape found in data assimilation ([Park and Zupanski, 2022](#)), implying our algorithm is working as expected.

Table S1: Internal climate variability noise parameters. Internal variability standard deviation ($\sigma_{q^{IV}}$) and AR(1) autocorrelation coefficient (ρ).

Noise model	$\sigma_{q^{IV}}$	ρ
AR(1)	0.27 ^a	0.20 ^b

^a From [Proistosescu and Huybers \(2017\)](#).

^b From [Bauer et al. \(2025\)](#).

Table S2: Regional noise model parameters. Internal variability magnitude ($\sigma_{q^{(r)}}$) for each region under the two-region and three-region configurations. Note each process is zero-mean AR(0).

Configuration	Region	$\sigma_{q^{(r)}}$
Two-region	Region 1	0.35
	Region 2	0.30
Three-region	Region 1	0.35
	Region 2	0.30
	Region 3	0.25

Table S3: Global prior distribution parameters. Prior means (μ) and standard deviations (σ) for global climate model parameters. Parameters whose prior σ is derived via the approach of Leach et al. (2021), where the mean is multiplied by a 30% error, are marked with #. The climate feedback parameter λ prior is set so that $ECS = 3.0$ when $F_{2\times}$ equals its average value, and its 5-95 percentile range matches Sherwood et al. (2020). The ocean heat transport coefficient is calibrated to result in a transient climate response prior that matches the IPCC estimate of 1.8 °C with a likely range of 1.4-2.2 °C.

Parameter	Symbol	μ	σ
Climate feedback ($\text{W m}^{-2} \text{K}^{-1}$)	λ	1.33	0.31
Ocean heat transport coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	γ	0.62	0.26
Pattern effect parameter (-)	ε	1.58	0.128 ^a
Surface heat capacity ($\text{J m}^{-2} \text{K}^{-1}$)	C_1	8.0	$0.3 \times \mu^\#$
Deep-ocean heat capacity ($\text{J m}^{-2} \text{K}^{-1}$)	C_2	100.0	$0.3 \times \mu^\#$
CO ₂ forcing (W m^{-2})	$F_{2\times}$	3.99	0.19 ^b

[#] $\sigma = 0.3\mu$ following Leach et al. (2021).

^a From Cummins et al. (2020).

^b From Smith et al. (2020).

Table S4: Regional prior distribution parameters. Prior means (μ) and standard deviations (σ) for pattern-scaling (α) and stratospheric aerosol injection (SAI) adjustment (β) parameters under the two-region and three-region model configurations.

Configuration	Parameter	Symbol	Region 1	Region 2	Region 3
Two-region	Pattern scaling (mean)	μ_α	1.6	1.3	—
	Pattern scaling (std)	σ_α	0.2816	0.2288	—
	SAI adjustment (mean)	μ_β	-0.1881	+0.1881	—
	SAI adjustment (std)	σ_β	0.1064	0.1064	—
Three-region	Pattern scaling (mean)	μ_α	1.6	1.3	1.275
	Pattern scaling (std)	σ_α	0.32	0.26	0.255
	SAI adjustment (mean)	μ_β	-0.193	+0.193	-0.119
	SAI adjustment (std)	σ_β	0.1484	0.1484	0.1484

Table S5: True (data-generating) global climate parameters. Fixed parameter values used to generate synthetic observations. Notation follows Table S3.

Parameter	Symbol	True value
Climate feedback ($\text{W m}^{-2} \text{K}^{-1}$)	$\lambda^\dagger$	1.33
Ocean heat transport coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	$\gamma^\dagger$	0.62
Pattern effect parameter (-)	$\varepsilon^\dagger$	1.58
Surface heat capacity ($\text{J m}^{-2} \text{K}^{-1}$)	$C_1^\dagger$	8.0
Deep-ocean heat capacity ($\text{J m}^{-2} \text{K}^{-1}$)	$C_2^\dagger$	100.0
CO ₂ forcing (W m^{-2})	$F_{2\times}^\dagger$	3.99

Table S6: True regional parameters, two-region configuration. Pattern-scaling ($\alpha^\dagger$) and SAI adjustment ($\beta^\dagger$) coefficients used in the data-generating model for each SAI angle ϑ (degrees).

$\vartheta^\dagger$ ($^\circ$)	Region 1		Region 2	
	$\alpha_1^\dagger$	$\beta_1^\dagger$	$\alpha_2^\dagger$	$\beta_2^\dagger$
5	1.6	-0.1754	1.3	-0.0247
10	1.6	-0.1881	1.3	+0.0727
15	1.6	-0.1881	1.3	+0.1689
20	1.6	-0.1881	1.3	+0.2579
25	1.6	-0.1881	1.3	+0.3417
30	1.6	-0.3188	1.3	+0.4010

Table S7: True regional parameters, three-region configuration. Pattern-scaling ($\alpha^\dagger$) and SAI adjustment ($\beta^\dagger$) coefficients for each SAI angle parameter ϑ .

$\vartheta^\dagger$ ($^\circ$)	Region 1		Region 2		Region 3	
	$\alpha_1^\dagger$	$\beta_1^\dagger$	$\alpha_2^\dagger$	$\beta_2^\dagger$	$\alpha_3^\dagger$	$\beta_3^\dagger$
5	1.600	-0.1927	1.300	-0.0175	1.275	-0.1193
10	1.600	-0.1927	1.300	+0.1024	1.275	-0.1193
15	1.600	-0.1927	1.300	+0.2128	1.275	-0.1193
20	1.600	-0.1927	1.300	+0.3171	1.275	-0.1193
25	1.600	-0.1927	1.300	+0.4172	1.275	-0.1193
30	1.600	-0.4167	1.300	+0.4894	1.275	-0.1193

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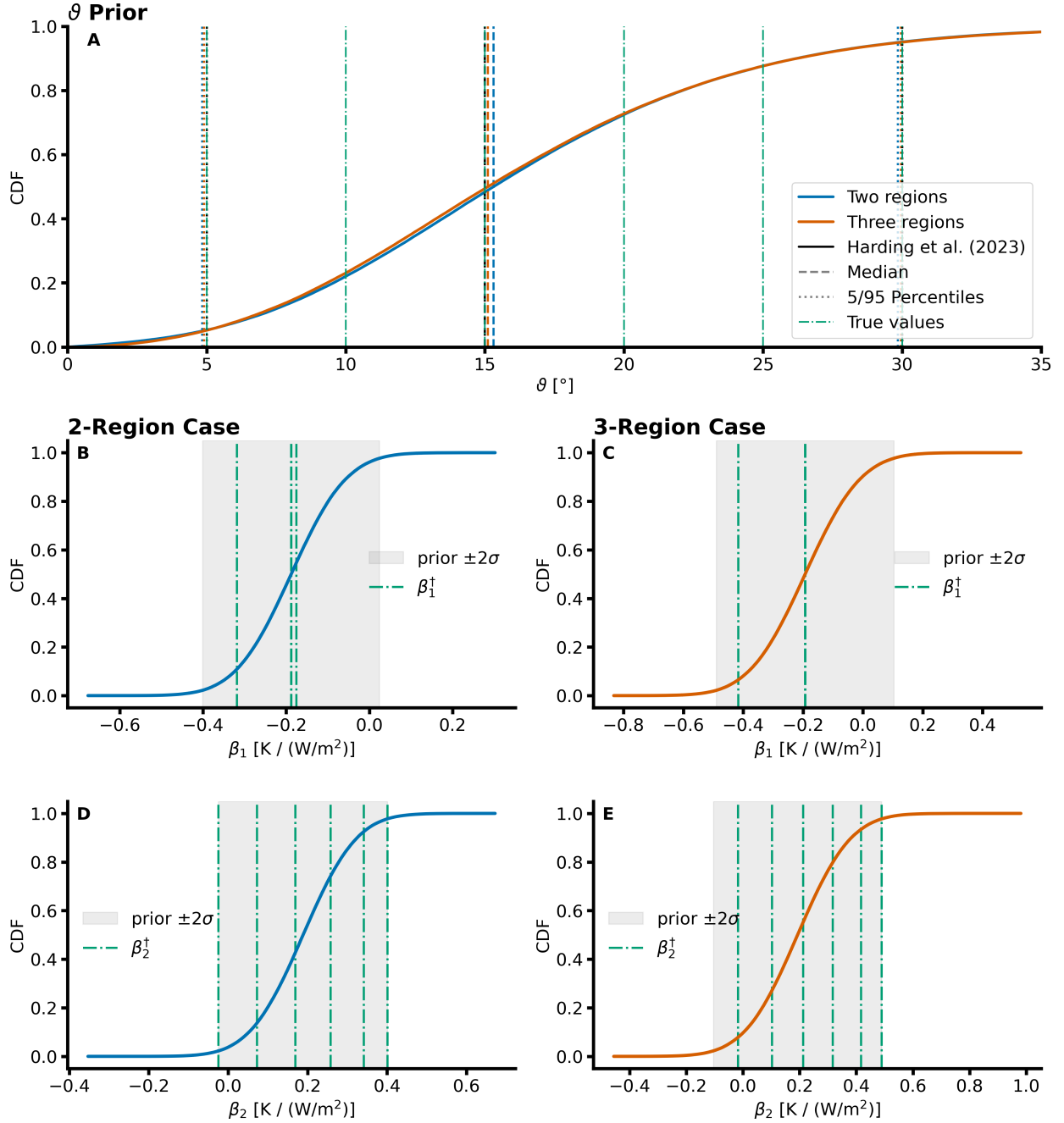


Figure S4: ϑ and β Priors. **A** shows the prior for ϑ for the two region (blue) and three region (orange) model. Colored vertical dashed lines show the prior median of each distribution, along with the [Harding et al. \(2023\)](#) reported median (black). Colored dotted vertical lines show the 5 and 95 percentiles. Green dashdot vertical lines show the true values of ϑ used in our simulations. **B** shows the prior for β_1 (black blue line) as well as the true values of β_1 (green dashdot vertical lines) used in the two region case. **C** is as **B** but for the three region model. **D** is as **B** but for β_2 . **E** is as **D** but for the three region model.

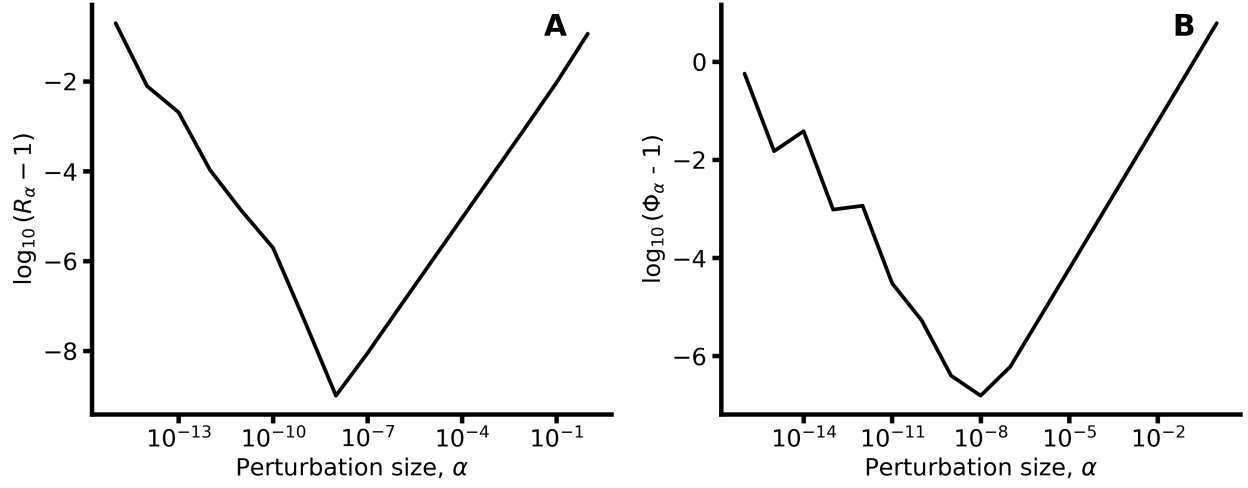


Figure S5: Tangent Linear Model and Cost Function Gradient Verification. **A** shows $\log_{10}(R_{\alpha} - 1)$, where R_{α} is given by (3.2). **B** shows $\log_{10}(\Phi_{\alpha} - 1)$ where Φ_{α} is given by (3.3). Each is plotted against the size of the perturbation, α . In both cases, the “V”-shape indicates that these model components are working as intended, see [Park and Zupanski \(2022\)](#).

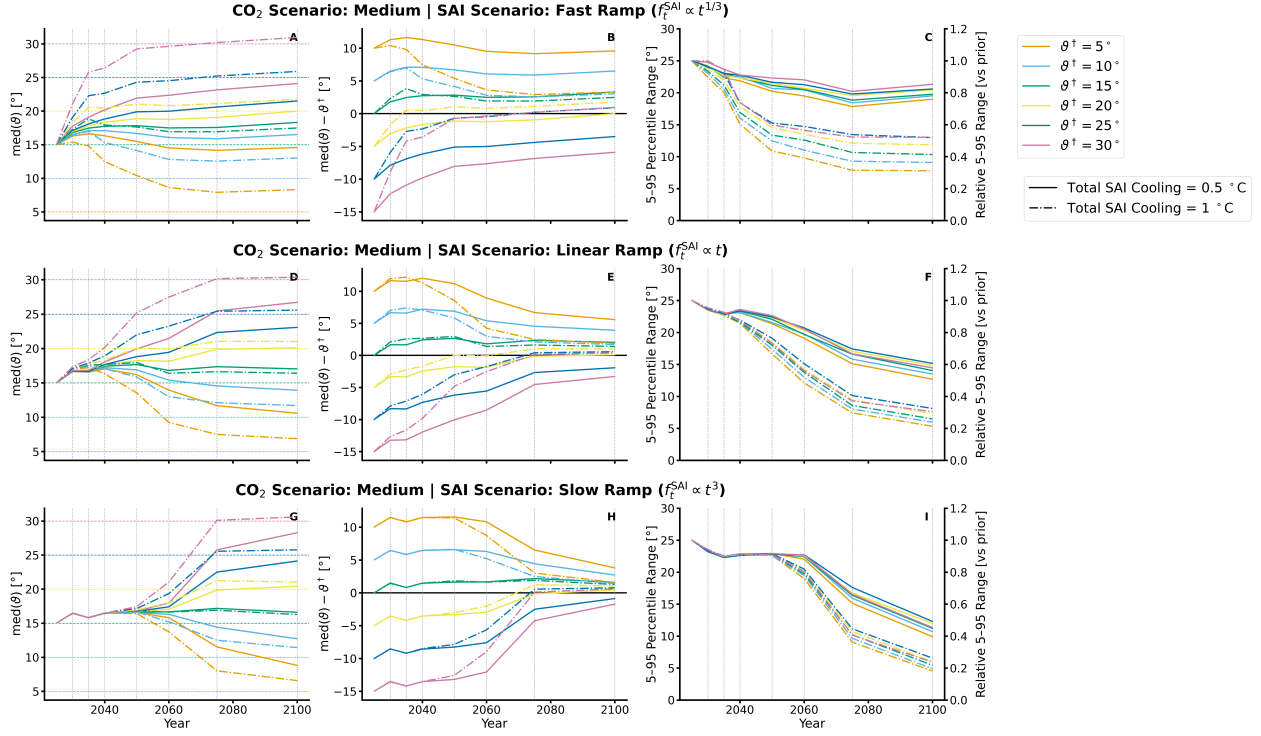


Figure S6: SAI Angle Learning Across SAI Scenario Designs, Three Region Model. **A** shows the ensemble median estimate of the SAI angle parameter ϑ as a function of time for the Medium CO_2 and fast SAI ramp scenario (see the row title). Dotted vertical lines indicate times when new observations are assimilated into the posteriors. The colors represent different true values of the angle parameter. Solid lines (dash-dot, resp.) indicate posterior medians when the total SAI cooling is $0.5\text{ }^\circ\text{C}$ ($1\text{ }^\circ\text{C}$, resp.). Dashed horizontal lines show the true value of the angle parameter for reference. **B** shows the bias, defined as the difference between the ensemble median estimate of the angle parameter and its true value, as a function of time. **C** shows the 5-95 percentile range in absolute terms (left axis) and relative to the prior 5-95 percentile range (right axis). **D–F** and **G–I** are as **A–C**, but for the linear SAI scenario and the slowly ramped up SAI scenario, respectively. Results follow the main text.

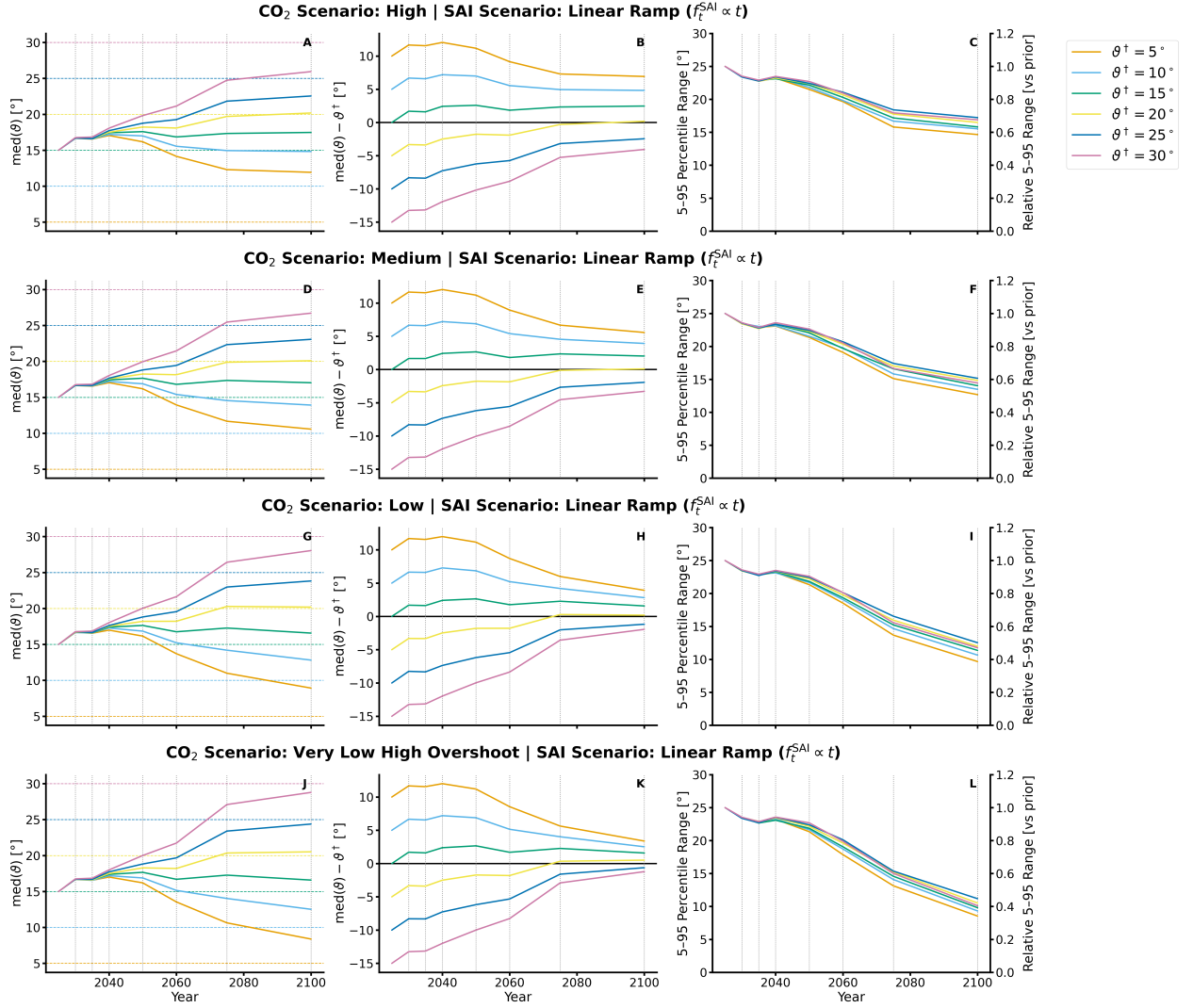


Figure S7: SAI Angle Learning Across CO₂ Scenarios, Three Region Model. **A** shows the ensemble median estimate of the SAI angle parameter ϑ as a function of time for the High CO₂ and linear SAI ramp scenario (see the row title). Dotted vertical lines indicate times when new observations are assimilated into the posteriors. The colors represent different true values of the angle parameter. Dashed horizontal lines show the true value of the angle parameter for reference. **B** shows the bias, defined as the difference between the ensemble median estimate of the angle parameter and its true value, as a function of time. **C** shows the 5-95 percentile range in absolute terms (left axis) and relative to the prior 5-95 percentile range (right axis). **D–F**, **G–I**, and **J–L** are as **A–C**, but for the Medium, Low, and Very Low High Overshoot scenarios, respectively. Results follow the main text.